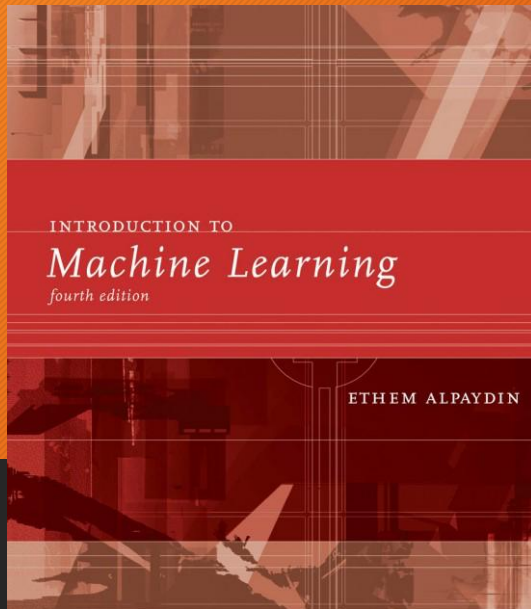




Lecture Slides for

INTRODUCTION TO Machine Learning

4e

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CHAPTER 1:

Introduction

Big Data

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- Widespread use of personal computers and wireless communication leads to “big data”
- We are both producers and consumers of data
- Data is not random, it has structure, e.g., customer behavior
- We need “big theory” to extract that structure from data for
 - (a) Understanding the process
 - (b) Making predictions for the future
- Cheaper computational power (e.g., GPUs).

Why “Learn” ?

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- Machine learning is programming computers to optimize a performance criterion using example data or past experience.
- There is no need to “learn” to calculate payroll
- Learning is used when:
 - Human expertise does not exist (navigating on Mars),
 - Humans are unable to explain their expertise (speech recognition)
 - Solution changes in time (routing on a computer network)
 - Solution needs to be adapted to particular cases (user biometrics)

What We Talk About When We Talk About “Learning”

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- Learning general models from a data of particular examples
- Data is cheap and abundant (data warehouses, data marts); knowledge is expensive and scarce.
- Example in retail: Customer transactions to consumer behavior:

*People who bought “Blink” also bought “Outliers”
(www.amazon.com)*

- Build a model that is *a good and useful approximation* to the data.

Data Mining

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- **Retail:** Market basket analysis, Customer relationship management (CRM)
- **Finance:** Credit scoring, fraud detection
- **Manufacturing:** Control, robotics, troubleshooting
- **Medicine:** Medical diagnosis
- **Telecommunications:** Spam filters, intrusion detection
- **Bioinformatics:** Motifs, alignment
- **Web mining:** Search engines
- ...

What is Machine Learning?

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- Optimize a performance criterion using example data or past experience.
- Role of Statistics: Inference from a sample
- Role of Computer science: Efficient algorithms to
 - Solve the optimization problem
 - Representing and evaluating the model for inference

Applications

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- Association
- Supervised Learning
 - Classification
 - Regression
- Unsupervised Learning
- Reinforcement Learning

Learning Associations

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- Basket analysis:

$P(Y | X)$ probability that somebody who buys X also buys Y where X and Y are products/services.

Example: $P(\text{chips} | \text{beer}) = 0.7$

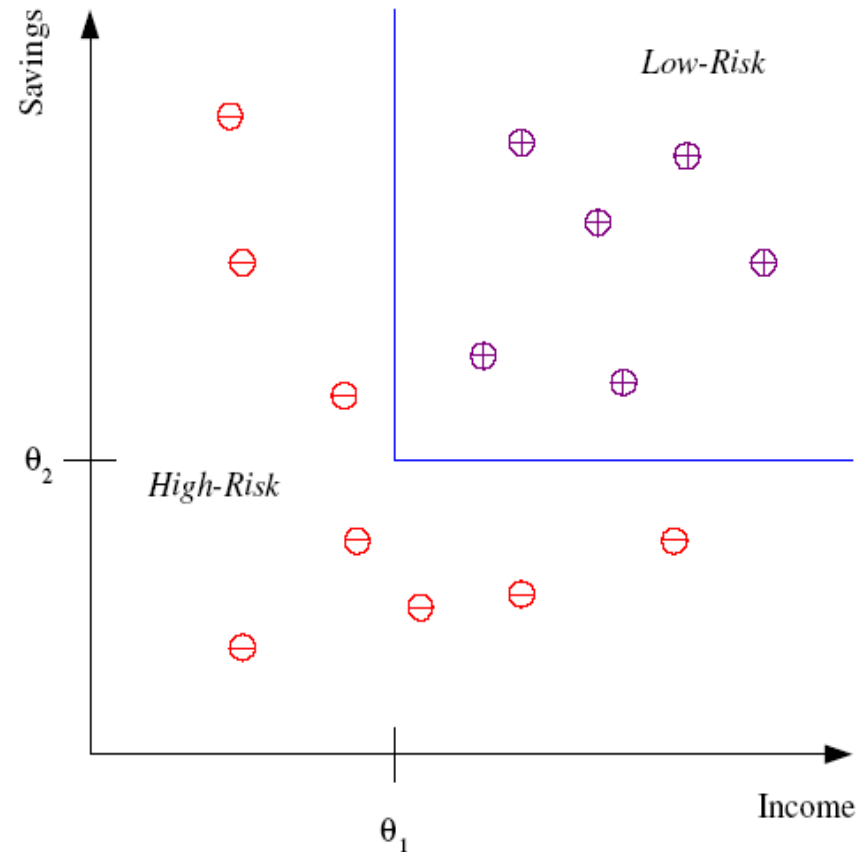
Classification

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- Example: Credit scoring
- Differentiating between low-risk and **high-risk** customers from their *income* and *savings*

Discriminant:

IF *income* $> \theta_1$ AND *savings* $> \theta_2$
THEN low-risk
ELSE **high-risk**



Classification: Applications

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- Aka Pattern recognition
- **Face recognition:** Pose, lighting, occlusion (glasses, beard), make-up, hair style
- **Character recognition:** Different handwriting styles.
- **Speech recognition:** Temporal dependency.
- **Medical diagnosis:** From symptoms to illnesses
- **Biometrics:** Recognition/authentication using physical and/or behavioral characteristics: Face, iris, signature, etc
- **Outlier/novelty detection:** Detecting faults

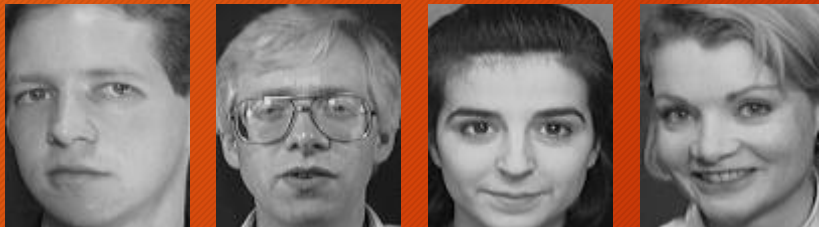
Face Recognition

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Training examples of a person



Test images



ORL dataset,
AT&T Laboratories, Cambridge UK

Regression

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- Example: Price of a used car

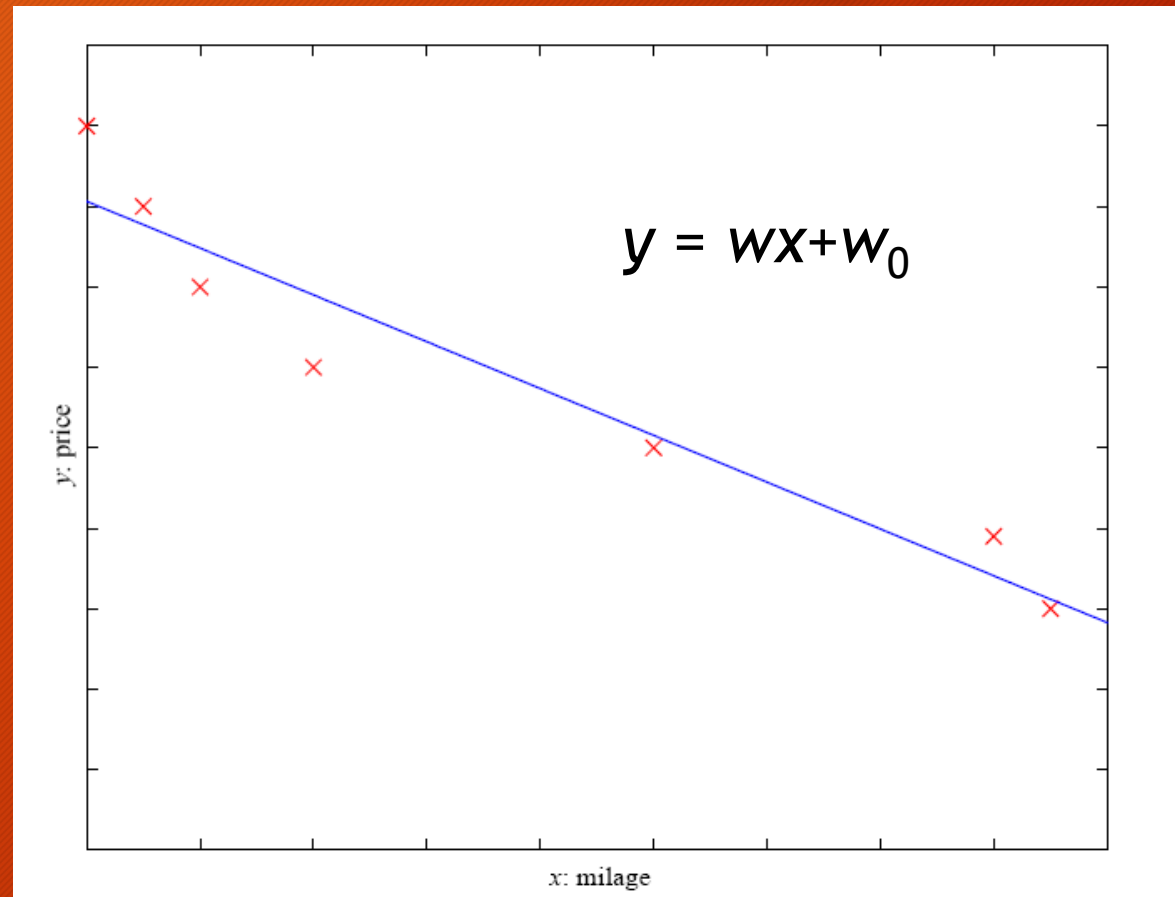
• x : car attributes

y : price

$$y = g(x | \theta)$$

$g()$, model (fixed)

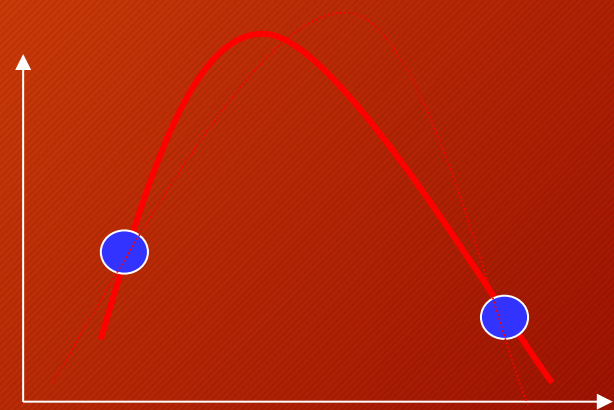
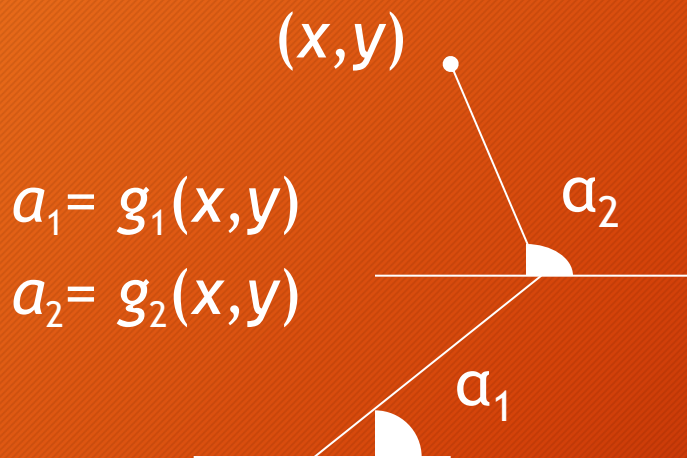
θ , parameters
(modifiable-
>learned)



Regression Applications

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- Navigating a car: Angle of the steering
- Kinematics of a robot arm



Response surface design

Supervised Learning: Uses

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- **Prediction of future cases:** Use the rule to predict the output for future inputs
- **Knowledge extraction:** The rule is easy to understand
- **Compression:** The rule is simpler than the data it explains
- **Outlier detection:** Exceptions that are not covered by the rule, e.g., fraud

Unsupervised Learning

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- Learning “what normally happens”
- No output
- Clustering: Grouping similar instances
- Example applications
 - Customer segmentation in CRM
 - Image compression: Color quantization
 - Bioinformatics: Learning motifs

Reinforcement Learning

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- Learning a **policy**: A **sequence** of outputs
- No supervised output but delayed reward
- Credit assignment problem
- Game playing
- Robot in a maze
- Multiple agents, partial observability, ...

- **High-performance computing:**
Parallel processing, GPUs
- **Data privacy and security:**
Sanitization/anonymization of personal data
privacy-preserving ML
- **Model interpretability and trust:**
The importance of testing
Bias in training data, fairness of decisions
Adversarial examples
We do not want black-box but explainable models

- A new field composed of
 1. **Machine learning** (theoretical component)
 2. **High-performance computation** (engineering component); both hardware and software aspects
 3. **Ethical/legal aspects** of data collection, analysis, and data-driven decision making (social component)